CYBERSECURITY FUNDAMENTALS

OSI - Open system Interconnection
Model to demonstrate the theory behind computer networking.
Practically the TCP/IP model (real networking)

OSI has 7 models:

(7) Applications: Provides networking options to programs running on computer.
Interface for transmit data
Passes down data to presentation layer.

(6) Presentation:
Recieves data from application layer
Application understandable format (not standard)
Translates data into a standardised format
(encryption, compression, or transformations).

(5) Session
Recieves data from presentation layer
- checks computer connection with other computer
- If connection not established → error
- If established, maintains & co-operates with remote comp.
- Unique to every question
Builds connection between host and remote computer

(4) Transport layer
- Transmits data over selected protocols

TCP (Transmisson Control Protocol)	UDP (User Data Protocol)
- Reliable transmisson	- High speed
- Slow but high accuracy	- Data might be lost
- bite sized pieces (segments)	- bite sized (datagrams)

③ Network:
⇒ Network layer takes IP address for information from a webpage
⇒ Logical addresses sort networks & categorise them.
IPV4 format.

② Data link (only layer to add both header & trailer)
Data from IP address of remote computer adder to MAC of recieving computer
MAC - (Media Access Control).
L set by manufacturer - cannot be changed
- Data link layer checks whether recieved data is not corrupted.

① Physical
- converts binary data into signals & transmit across network.

Encapsulation - Process by which data can be sent from one computer to another
De - encapsulation - reverse process, starts at the physical layer to application layer

data is recieved & transmitted

Data at various OSI layers.

- L7 - Data.
- L6 - Data
- L5 - Data
- L4 - Data (Segments/ Datagrams)
- L3 - Data (Packets)
- L2 - Data (Frames)
- L1 - Data stream (Bits)

• Frames used for error detection
Encapsulation provides extra layer of security

TCP/IP model

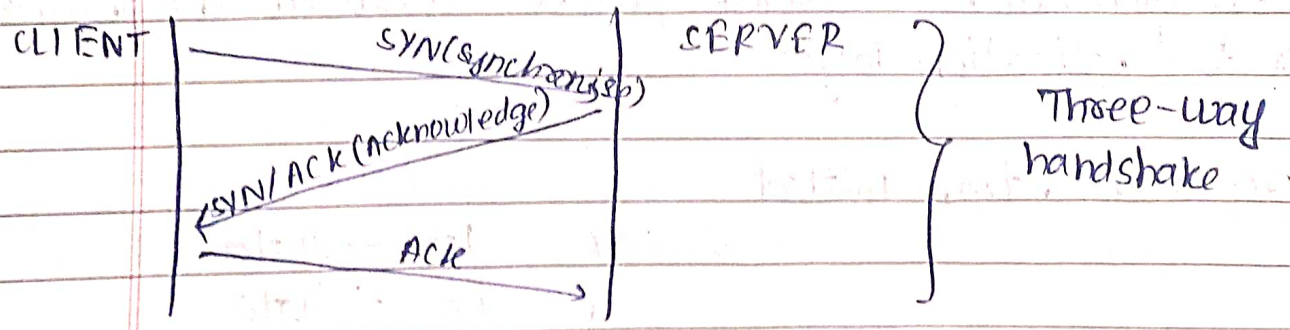
- Similar to OSI model
- Basis of real world networking
  OSI — more rigid & less condensed than TCP/IP model.

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	Network Interface
Physical	

- Encapsulation & de-encapsulation works similar.
  - Encapsulation → header adder
  - de-encapsulation → header removed.

TCP - Transmission Control Protocol - controls flow of data
IP - Internet Protocol - how data packets are sent

TCP (connection based protocol)
- Forms stable connection b/w computers - 3 way handshake



A strong & permanent connection.

→ TCP/IP models introduced in 1982 for standardisation for networking.

DNS (Domain name system)

- DNS gives the IP address of the website
- First request is sent to DNS server & then the server would look for IP address for website

(I) computer checks local hosts file if IP->Domain mapping has been created.
Checks local DNS for any IP address stored earlier

(II) If no IP address found, computer sends request to recursive DNS server.
details for rDNS are stored in router or computer
- RDNS maintains cache for popular domains; if not fount passes to ~~root~~ name server

- (III) Root name servers chooses DNS servers next level down i.e. Top-level Domain servers according to request

(IV) TLD servers - works according to extensions.
.com domain TLD servers accept website requests with .com extension.

(V) Authoritative name servers - stores DNS records of all domains
When request reaches to authoritative name server, gives relevant information & gives IP address.

<u>dig</u> - manually query recursive DNS server for choice.

dig <domain> @dns-server-ip>
TTL - tells computer when it should request data again (when record expires).
Two google public DNS servers - 8.8.8.8
8.8.4.4